End Course Summative Assignment

1. **What is a vector in mathematics?**

   A vector is a 1D array. Vector is a mathematical quantity with both magnitude and direction. The magnitude of vector u, denoted |u|, is the length. The way to express its direction is to give the angle it makes with a horizontal ray that is parallel to the positive x-axis. This is called the vector's direction angle. For instance, a point in space can be defined as a vector of three coordinates (x, y, z).

2. **How is a vector different from scalar?**

Scalar	**Vector**
A scalar has only magnitude.	A vector has both magnitude and a direction.
Scalars use a single number and a unit (Eg., 5 Kg).	Vectors use a number, a unit, and a pointing direction (Eg., 5 m/s North).
Scalars can be added, subtracted, increased, and divided using the usual arithmetic operations.	Inaddition to the usual arithmetic operations, vectors use geometric rules that include angles and directions.
Scalars are used to represent physical amounts that have size but no direction, such as temperature, mass, or time.	Vectors are utilized to represent physical quantities that have both measure and direction, such as velocity, force, or displacement.

3. **What are the different operations that can be performed on vectors?**

   Given two vectors u and v in a vector space, vector operations are defined as.

Vector Operation	**Formula**
Vector Addition (u+v)	$(u+v)_i = u_i + v_i$ for i=1, 2,..., n
Vector Subtraction (u-v)	$(u-v)_i = u_i - v_i$ for i=1, 2,..., n
Scalar Multiplication (cu)	$(cu)_i = c*u_i$ for i=1,2,...,n
Dot Product (u.v)	$u.v = u_1v_1 + u_2v_2 + ... + u_nv_n$
Cross Product (u x v)	$u \times v = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$

Vector properties for vector operations are as following.

	Properties							
Vector Operation	**Commutative**	**Associative**	**Distributive**	**Identity**	**Inverse**	**Zero**	**Orthogonality**	**Parallelism**
Vector Addition (u+v)	✅	✅		✅	✅			
Vector Subtraction (u-v)								
Scalar Multiplication (cu)		✅	✅	✅		✅		
Dot Product (u.v)	✅	✅	✅				✅	
Cross Product (u x v)			✅				✅	

4. How can vectors be multiplied by scalar?

Multiplying a vector a by a constant scalar c gives a vector whose direction is the same, but the magnitude is changed by a factor of c. Scalar Multiplication of a vector u is represented as,

$$|cu| = c|u|$$

If c > 1, the magnitude of the vector increases, while it decreases when c < 1.

Example, *u = (1, 2, 3) and c = 2.*

$c \cdot u = 2 \cdot (1,2,3) = (2, 4, 6)$.

Properties of scalar multiplication:

Distributivity: $c(u+v) = cu + cv$

Associativity: $(c_1c_2)u = c_1(c_2u)$

Identity element: $1u = u$ for all vectors u

Zero element: $0u = 0$ for all vectors u

5. What is the magnitude of a vector?

The magnitude of vector u, denoted $|u|$, is the length and is always a positive scalar value. It represents the distance from the origin to the point (x, y, z).

For the given vector $\bar{A} = xi + y\hat{j} + z\hat{k}$, then the magnitude of vector $\bar{A}$ can be calculated using the Pythagorean theorem. It takes the square root of the sum of the squares of its components.

$|A| = \sqrt{x^2 + y^2 + z^2}$

6. How can the direction of a vector be determined?

The way to express direction of a vector is to give the angle it makes with a horizontal ray that is parallel to the positive x-axis. This is called the vector's direction angle.

For a vector a with horizontal component x and vertical component y, the slope is $\theta = \tan^{-1}(y/x)$

The direction of the vector θ is computed based on the quadrant (x, y).

Quadrant in which (x, y) lies	**θ (in degrees)**
1	θ
2	180° - θ
3	180° + θ
4	360° - θ

7. What is the difference between a square matrix and a rectangular matrix?

Square matrices: A square matrix has an equal number of rows and columns.

Rectangular matrices: A rectangular matrix has an unequal number of rows and columns.

8. What is a basis in linear algebra?

A basis in linear algebra is a set of vectors that acts as a foundational framework for a vector space and represent every vector in a vector space. For a set of vectors to be a basis, it must satisfy two conditions:

Vectors span the entire vector space: Every vector in the space can be written as a linear combination of the basis vectors.

Linearly Independent: No vector in the set can be written as a combination of the others.

9. What is a linear transformation in linear algebra?

Linear transformation defines how points or vectors are transformed while preserving the linear properties of the space such as vector addition and scalar multiplication. A transformation T is linear if, for all vectors u, v and scalars c, $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$. These properties can be combined into one is represented as $T(au + bv) = aT(u) + bT(v)$, for any scalars a and b. It is used to perform operations such as rotation, scaling, reflection, and shearing by multiplying the matrix with a vector. Some common types of transformation matrices include translation matrix, rotation matrix, scaling matrix, reflection matrix, shear matrix, affine transformation matrix, and combined matrix.

10. What is an eigenvector in linear algebra?

Eigenvectors are non-zero vectors that, when a linear transformation such as a matrix multiplication is applied to it, change only in magnitude, not direction. It is only scaled, stretched, or shrunk by a factor called an eigenvalue. For an n × n matrix A, eigenvectors are n × 1 column vectors. The eigen vector equation is given as,

$$Av = \lambda v$$

where,

- *A is the given square matrix.*
- *v is the eigenvector of matrix A. It must not be a zero vector.*
- *λ is scalar multiple. If λ* is positive, the vector stretches; if negative, it changes direction**.**

It is also represented as

$$(A - \lambda I)x = 0$$

where, I is the identity matrix.

This means that the eigenvectors of A correspond to the non-trivial solutions of the homogeneous linear system $(A - \lambda I)x = 0$. The set of all eigenvalues of A is called the spectrum of A.

11. What is the gradient in machine learning?

Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is a way to find the best parameters for the model by iteratively moving towards the minimum of the cost function. Gradient shows how the error changes when we change the model parameters. The gradient points in the direction of greatest increase in loss. So, the parameters are moved in the direction opposite to the gradient to reduce the loss. Gradient descent is given by the formula,

$$w - \eta \nabla L$$

where,

- w = model parameters
- ∇L = gradient of the loss
- η = learning rate

The gradient is positive, meaning increasing w would increase the loss. The negative gradient points in the direction of greatest decrease in loss. Gradient descent therefore moves w in the negative direction.

12. What is backpropagation in machine learning?

In a neural network, the process of adjusting the weights and biases to minimize the difference between the actual output and the desired output is known as backpropagation. It works by propagating errors backward, computing gradients of losses for each parameter using the chain rule, and updating weights and biases to improve performance. Backpropagation for the model $\hat{y} = w_1 x_1 + w_2 x_2 + b$ is given by:

- Forward Pass: The input gives the prediction $\hat{y}$.
- Error Calculation: Compare the prediction $\hat{y}$ with the actual value y using a loss function $L = (y - \hat{y})^2$.

- Back Propagation: Computes gradients of the loss function with respect to each weight using the chain rule.

$$\frac{\partial \mathrm{L}}{\partial \mathrm{w}_1}, \frac{\partial \mathrm{L}}{\partial \mathrm{w}_2}, \frac{\partial \mathrm{L}}{\partial \mathrm{w}_3}$$

  The derivatives tell us how changing each weight would change the loss.
- Update the weights: The weights and biases are then updated to reduce the error.

$$w_{new} = w_{old} - \eta \frac{\partial \mathrm{L}}{\partial \mathrm{w}} .$$

13. What is the concept of a derivative in calculus?

Differentiation is the process of finding the derivative of a function. A derivative in calculus is the rate of change of a quantity y with respect to another quantity x. It represents the change of function at a given point. The derivative of a function at a point is the slope of the tangent drawn to the curve at the point.

The derivative of a function f(x) at a point x=a is given by f'(a) and is characterized as the limit of the difference quotient as the distinction between x and a approaches zero.

$$f'(a) = lim\,[(f(x) - f(a))/(x - a)] \text{ as x approaches a.}$$

Derivatives are also used to find the intervals where the function is increasing/decreasing as well as the intervals where the function is concave up/down.

14. How are partial derivatives used in machine learning?

In machine learning, models have many parameters like weights and biases. The loss function shows how far predictions are from the correct output. To improve the model, we need to know how each parameter affects the loss. Partial derivatives help us update each parameter individually, which is the idea behind gradient descent. By measuring how the loss changes with respect to each parameter, we adjust them to reduce error.

15. What is probability theory?

Probability theory is the branch of mathematics that deals with uncertainty and randomness. Probability theory, defines events as subsets of a sample space, which is the set of all possible outcomes of a random experiment. It assigns a numerical measure to the likelihood of each event occurring. Probability is always a number between 0 and 1, with 0 indicating that an event is impossible and 1 indicating that an event is certain.

16. What are the primary components of probability theory?

Experiment: An experiment is a trial for the likelihood of an event occurring. Example: Tossing a coin.

Sample Space: A sample space is the set of all possible outcomes of an experiment. Example: Sample Space = {Heads, Tails}

Outcome: An outcome is the result of experiment. Example: {Heads}

Event: An event is a collection of outcomes. Example: {Heads, Tails}

Probability Measure: It is a number between 0 and 1 to each event, showing how likely it is to happen. If an event is impossible, its probability is 0. If an event is certain, its probability is 1.

17. What is conditional probability and how is it calculated?

Conditional probability is the probability of an event happening given that another event has already occurred.

Conditional Probability for dependent events is given as,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where, $P(B) > 0$

$P(A|B)$ represents the probability of event A given event B has occurred.

$P(A \cap B)$ represents the probability of both events A and B occurring simultaneously.

$P(B)$ represents the probability of event B occurring.

Conditional Probability for independent events is given as,

$$P(A|B) = P(A)$$

where,

$P(A)$ represents the probability of event A occurring.

18. What is Bayes theorem and how is it used?

Bayes' Theorem is a statistical formula that allows us to calculate the probability of an event occurring based on prior knowledge or evidence. It allows us to update the probability of an event based on new evidence. It states that the probability of an event A given that another event B has occurred can be calculated as

$$P(A|B) = P(B|A) * P(A) / P(B)$$

where,

P(A|B) is the probability of event A given that event B has occurred

P(B|A) is the probability of event B given event A has occurred

P(A) is the prior probability of event A

P(B) is the prior probability of event B.

19. What is a random variable and how is it different from regular variable?

Regular Variable	**Random Variable**
Represents a value that can change or is unknown	Represents the numerical outcome of a random experiment
Value can be assigned directly	Value is determined by a random outcome
Does not necessarily involve probability	Has probabilities associated with its possible values
Example: x=10	Example: X= result of a die roll

20. What is the law of large numbers, and how does it relate to the probability theory?

The law of large numbers describes the relationship between the sample size of a random variable and its expected value. This law applies to both discrete and continuous random variables. The law of large numbers states that as the sample size of a random variable increases, the sample mean will approach the expected value of the variable.

For random variables X_1, X_2, ... X_n with mean μ and variance σ^2, the sample mean $\bar{X}_n = (X_1 + X_2 + ... + X_n) / n$ converges in probability to the true mean μ. This means that, for any $\varepsilon > 0$,

$$lim\ P(|X_n\ -\ \mu|\ >\ \varepsilon)\ =\ 0$$

21. What is the central limit theorem and how is it used?

The central limit theorem (CLT) states that the distribution of the sample mean of a large number of independent and identically distributed random variables approaches a normal distribution, regardless of the underlying distribution of the individual random variables. The central limit theorem holds under certain conditions, including that the sample size is sufficiently large, the observations are independent, and the population distribution has a finite variance.

Let X_1, X_2, ..., X_n be a sequence of random variables with mean μ and variance σ^2. Then, the sum of these random variables, $S = X_1 + X_2 + ... + X_n$, will have a distribution that approaches a normal distribution as n grows large. Specifically, as $n \to \infty$,

$$(S\ -\ n\mu)\ /\ (\sigma\sqrt{n}) \to N(0, 1)$$

where N(0, 1) denotes a standard normal distribution with mean 0 and variance 1. It means that, for large n, we can use the standard normal distribution to approximate the distribution of S.

22. What is the difference between discrete and continuous probability distributions?

Feature	Discrete Probability Distribution	Continuous Probability Distribution
Data Type	Distinct, countable values (e.g., (0, 1, 2, 3))	Any value within an interval (e.g., (1.5, 3.14))
Function Used	The probability function for a discrete random variable is the probability mass function**.** Gives the exact probability of a specific value.	The probability function for a continuous random variable is the Probability Density Function. Measures the relative likelihood of a value falling within a specific range.
Single Value Probability	$P\ (X{=}x\ )$	($P\ (X{=}x\ )$= 0) Probabilities are only calculated for ranges.
Total Probability Rule	$\sum P(X)$ = 1	$\int f(x)dx$ = 1
Types	Binomial, Poisson, Geometric	Normal (Gaussian), Uniform, Exponential

23. What are some common measures of central tendency, and how are they calculated?

Central Tendency identifies the central position within that set of data. The common measures of central tendency are as below.

- Mean: The mean is equal to the sum of all the values in the data set divided by the number of values in the data set. It is susceptible to outliers when unusual values are added it gets skewed i.e deviates from the typical central value.Mean of as set of variables is given as,

$$\bar{x} = \frac{x_1 + x_2 + \cdots x_n}{n}$$

- Median: It is the middle value in a dataset when the values are arranged in ascending or descending order. It is less affected by

outliers and skewness of the data. The median value is much closer than the typical central value. Index of the median of n variables is given as,

$$n_{even} = \left(\frac{n+1}{2}\right)$$

$$n_{odd} = \frac{n}{2}$$

- Mode: It is the most frequently occurring value in a dataset. Mode can be used for both numerical and categorical data. A dataset can have one or more modes, or it may have no mode at all.

24. What is the purpose of using percentiles and quartiles in data summarization?

Percentiles and quartiles are used to divide a dataset into equal parts. Percentiles and quartiles can be calculated using various methods, such as interpolation, nearest rank, and the inverse of the cumulative distribution function.

Percentiles divide a dataset into 100 equal parts, each representing 1% of the data.

Quartiles divide a dataset into four equal parts, each representing 25% of the data. The first quartile (Q1) is the 25th percentile, the second quartile (Q2) is the 50th percentile (Median), and the third quartile (Q3) is the 75th percentile.

25. How do you detect and treat outliers in a dataset?

<u>Outlier Detection:</u>

Outlier detection is a process of identifying observations or data points that significantly deviate from the majority of the data. Outlier can be detected either by statistical methods or by visualization

methods. The statistical methods of outlier detection includes Z-score, IQR method.

- Z-score: Measures how many standard deviations a point is from the mean. Z-score threshold is set, commonly 3, and any data point with a Z-score exceeding this threshold is considered an outlier.

$$z = \frac{x - \mu}{\sigma}$$

where, z = Z-score, x = Value being evaluated (raw score),

μ = Mean of the population, σ = Standard deviation of the population.

- IQR Method (Interquartile Range**):** It calculates the IQR as the difference between the (Q3)75th and (Q1)25th percentiles of the data. Outliers lie outside the upper and lower bound values.

IQR = Q3-Q1

Lower bound: Q1–1.5 × IQR

Upper bound: Q3 + 1.5 × IQR

The visualisation method of Outlier detection includes Box Plot, Scatter Plot, Histogram/Density Plot, Pair Plot.

- Box Plot: Spot outliers beyond whiskers.
- Scatter Plot: Shows outliers in relation to features.
- Histogram: Identifies long tails or unusual spikes.
- Pair Plot: Identifies outliers across feature combinations.

Outlier Treatment:

Outlier Treatment involves treating detected outliers. Common methods of outlier treatment include Trimming, Winsorization, Imputation.

- Trimming: Drop rows with extreme values with data entry errors or make up a tiny fraction (<5%) of the dataset.
- Winsorization: Replace values outside the lower and upper bounds or above and below a nth percentile value with the threshold limit itself.
- Imputation: Treat outliers as missing values and fill them using the median or a k-Nearest Neighbours imputer.

26. How do you use a central limit theorem to approximate the discrete probability distribution?

27. How do you test the goodness of fit of a discrete probability distribution?

Goodness of fit is a statistical measure that indicates how well a model explains or represents the observed data. A good fit means that the model's predictions are close to the actual observations.

Chi-Square Goodness of Fit Tests: Chi-square test compares observed frequencies (from data) with expected frequencies (from hypothesis). It is used to test how well the data fits an expected distribution. Assumptions of the Chi-squared test includes,

1. Random sample: Data randomly collected
2. Independent observations: Each observation independent
3. Expected frequencies:
    - At least 80% of categories have expected frequency ≥ 5
    - No more than 20% of categories have expected frequency < 5
    - No category has expected frequency < 1

The formula for Chi-Square test statistic χ^2 is given by,

$$\chi^2 = \sum_{i=1}^{n} \frac{(O-E)^2}{E}$$

where, O is the observed value

E is the expected value $E = \frac{n}{k}$ (uniform distribution) and

$E = n \times p$ (specific distribution)

n – number of categories

df (degrees of freedom) = n - 1

Hypotheses Testing is performed as follows,

- H_o (null hypothesis): The data follow the hypothesized distribution
- H_1 (alternate hypothesis): The data do not follow the hypothesized distribution

Decision based on χ^2_critical value,

If $\chi^2 > \chi^2$_critical: Reject H_o (null hypothesis)

If $\chi^2 \leq \chi^2$_critical: Fail to reject H_o (null hypothesis)

Decision based on p-value for significance level α (typically 0.05),

If p-value $< \alpha$: Reject H_o (null hypothesis)

If p-value $\geq \alpha$: Fail to reject H_o (null hypothesis)

28. What is a joint probability distribution?

Joint Probability Distribution describes the probability of two or more random variables taking on specific values simultaneously. The variables take on specific values (if they are discrete) or fall within a region (if they are continuous). They allow us to model the relationships between variables and calculate the probability of specific events occurring. For X and Y are two random variables, it gives the probability $X = x$ and $Y = y$ occurring together.

- The joint probability distribution for two continuous random variables X and Y is represented by the joint probability density function (Joint PDF).

$$f_{X,Y}(x, y)$$

which describes the density of probability over the (x, y) plane.

- The joint probability distribution for discrete random variables, it is given by the joint probability mass function (Joint PMF).

$$P(X = x,\ Y = y)$$

Joint distributions have several important properties, including independence, covariance, and correlation, non-negativity and the integral over the entire domain equals to one. Some common joint distributions include the bivariate normal distribution and the joint uniform distribution.

29. How do you calculate the joint probability distribution?

Based on Nature of Random Variable:

The joint probability distribution function $P(X = x, Y = y)$ for X and Y two random variables are given based on the continuous or discrete nature of the variable.

- The joint probability distribution for two continuous random variables X and Y is represented by the joint probability density function (Joint PDF).

  $$f_{X,Y}(x, y)$$

  which describes the density of probability over the (x, y) plane.

- The joint probability distribution for discrete random variables, it is given by the joint probability mass function (Joint PMF).

  $$P(X = x,\ Y = y)$$

Based on Nature of Event:

Joint Probability is calculated based on whether the event is dependent or independent.

- Independent events: The occurrence of one event does not affect other. Joint probability is calculated using the multiplication rule.

  $$P(X = x, Y = y) = P(X = x)P(Y = y)$$

  where, $P(X = x)$ is the probability of occurrence of event X,

$P(Y = y)$ is the probability of occurrence of event Y,
$P(X = x, Y = y)$ is the joint probability of events $X = x$ and $Y = y$.

- Dependent events: The occurrence of one event affects probability of other. The joint probability uses conditional probability.

$$P(X = x, Y = y) = P(Y = y \mid X = x)P(X = x)$$

where, $P(X = x)$ is the probability of occurrence of event X,
$P(Y = y \mid X = x)$ is the conditional probability of occurrence of event Y when event X has already occurred,
$P(X = x, Y = y)$ is the joint probability of events $X = x$ and $Y = y$.

30. What is the difference between a joint probability distribution and a marginal probability distribution?

31. What is the covariance of a joint probability distribution?

Covariance measures the degree to which two random variables X and Y are linearly related. It is defined as the expected value of the product of the deviations of X and Y from their respective means.

$$cov(X, Y) = E[(X - E[X])(Y - E[Y])]$$

32. How do you determine if two random variables are independent based on their joint probability distribution?

Joint distributions have several important properties, including independence, covariance, and correlation. Two random variables X and Y are independent if and only if the joint density function $f_{X,Y}(x, y)$ can be expressed as the product of the marginal densities $f_X(x)$ and $f_Y(y)$.

$$f_{X,Y}(x, y) = f_X(x) * f_Y(y)$$

33. What is the relationship between the correlation coefficient and the covariance of a joint probability distribution?

Covariance and correlation measure the strength of the linear relationship between two random variables.

Covariance measures the degree to which two random variables X and Y are linearly related. It is defined as the expected value of the product of the deviations of X and Y from their respective mean. The covariance of X and Y is defined as,

$$Cov(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Correlation is a standardized version of covariance, and measures the degree of linear association between two variables X and Y. Correlation coefficient is related to covariance as,

$$\rho(X, Y) = Cov(X, Y) / (\sigma_X * \sigma_Y)$$

where, σ_X and σ_Y are the standard deviations of X and Y, respectively.

34. What is sampling in statistics, and why is it important?

Data sampling is a statistical method that selects a representative subset (sample) from a large dataset. The significance of data sampling are as listed below.

- **Cost and Time Efficient:** Sampling reduces time and resources needed by analysing a portion of the data instead of entire dataset.
- **Feasible for Large Populations:** When it is too costly or impractical to study the entire or infinite population, sampling is the only manageable and effective solution.
- **Reduces Risk of Error:** By using proper sampling methods, researchers can avoid biases and minimize the influence of outliers.
- **Maintains Accuracy:** A well-chosen sample can accurately reflect the homogeneous nature of the larger population when testing.
- **Reliability:** If samples are taken in proper size and on proper grounds the results of sampling will be reliable and representative of the source dataset.

- **Scientific:** The methods used to collect data are based on scientific theory and results obtained can be tested, sampling is a more scientific method of collecting data.

35. What are the different sampling methods commonly used in statistical inference?

Data sampling could be classified as two main techniques Probability Data Sampling and Non-Probability Data Sampling.

Probability Data Sampling Technique:

Probability Sampling ensures every data point has a known, non-zero chance of being selected. This helps create a representative sample, allowing reliable generalization to the whole population.

- **Simple Random Sampling**: Each data point has an equal chance of selection. For example, tossing a coin head or tail has equal probability.
- **Systematic Sampling**: Data is selected at regular intervals from an ordered list. For example, from 10 entries, selecting every 2nd one (2nd, 4th, 6th.).
- **Stratified Sampling**: Data is divided into groups (strata) based on shared traits and samples are drawn from each group. For example, dividing employees by gender, age, income and then sampling from each group.
- **Cluster Sampling**: Data is divided into random groups (clusters) and then subset sampled within each cluster. For example, choosing random user groups from different mobile networks.

Non-Probability Data Sampling Technique:

Non-probability data sampling means that the selection happens on a non-random basis and it depends on the individual as to which data does it want to pick. There is no random selection and every selection is made by a thought and an idea behind it.

- **Convenience Sampling**: Data is selected based on ease of access and minimal effort. For example, choosing the most recent or easily available resource data.
- **Voluntary Response Sampling**: Participants choose to take part in the study on their own. For example, social awareness survey where only willing participants respond.
- **Purposive Sampling**: Data is selected for a specific purpose or characteristic. For example, surveying rural areas to study educational needs.

36. What is the central limit theorem, and why is it important in statistical inference?

The central limit theorem (CLT) states that the distribution of the sample mean of a large number of independent and identically distributed random variables of the same size from a population approaches a normal distribution, regardless of the underlying distribution of the individual random variables. The central limit theorem holds under certain conditions, including that the sample size is sufficiently large, the observations are independent, and the population distribution has a finite mean and variance.

Let $X_1, X_2, ..., X_n$ be a sequence of random variables with mean μ and variance σ^2. Then, the sum of these random variables, $S = X_1 + X_2 + ... + X_n$, will have a distribution that approaches a normal distribution as n grows large. Specifically, as $n \to \infty$,

$$(S - n\mu) / (\sigma\sqrt{n}) \to N(0, 1)$$

where N(0, 1) denotes a standard normal distribution with mean 0 and variance 1. It means that, for large n, we can use the standard normal distribution to approximate the distribution of S.

The Central Limit Theorem is important because it provides the theoretical foundation for making inferences about a population from a sample. It allows us to,

1. Estimate population parameters — We can use a sample mean to estimate the population mean.
2. Construct confidence intervals — The approximate normality of the sample mean allows us to calculate confidence intervals for population parameters.
3. Perform hypothesis tests — Many statistical tests rely on the sampling distribution being approximately normal.
4. Understand sampling variability — The standard error $\sigma \,/\, \sqrt{n}$ shows how estimates become more precise as sample size increases.

37. What is the difference between parameter estimation and hypothesis testing?

38. What is the p-value in hypothesis testing?

P-value is a measure used in statistical testing to determine the strength of the evidence of observed data against a null hypothesis. P-value measures the probability of obtaining the observed results, assuming that the null hypothesis is true.

Statistical significance is determined by comparing the p-value to a chosen alpha (α) or significance level. A smaller p-value suggests stronger evidence against the null hypothesis and vice versa. The decision based on p-value for significance level α is based on,

- If p-value < α: Reject H_o (null hypothesis)
- If p-value ≥ α: Fail to reject H_o (null hypothesis)

The significance level of various α threshold is given below,

α=0.05 threshold	Evidence Against H_o	Decision
p≤0.001	Very Strong	Reject H_o
0.001 ≤ p < 0.01	Strong	Reject H_o
0.01 ≤ p < 0.05	Moderate	Reject H_o
0.05 ≤ p < 0.1	Weak	Fail to reject H_o
p ≥ 0.1	Very Weak	Fail to reject H_o

39. What is the confidence interval estimation?

Confidence intervals are a range of values within which we can be confident that the true population parameter lies. This range is estimated based on a sample from the population and a chosen level of confidence. The level of confidence speaks to the likelihood that the genuine populace parameter lies inside the certainty interim.

$$Confidence\ Interval = sample\ mean \pm (critical\ value)(standard\ error)$$

where,

The critical value is based on the level of certainty and the degrees of freedom (df) from a t-distribution or z-distribution.

Degrees of freedom(df) = n-1 for a test of measure n.

Standard Error = σ/ √(n).

40. What are Type-I & Type-II errors in hypothesis testing?

Type I error is a false positive result in hypothesis testing. It happens when we fail to dismiss the invalid speculation, even though it is wrong. The likelihood of making a Type I error is signified by the Greek letter alpha (α) and depends on the test estimate, impact estimate, and centrality level.

Type II error is a false negative result in hypothesis testing. It occurs when we fail to reject the null hypothesis, even though it is false. The probability of making a Type II error is denoted by the Greek letter beta (β) and depends on the sample size, effect size, and significance level.

41. What is the difference between correlation & causation?

Correlation	**Causation**
Correlation is the strength and direction of a (linear) relationship	Causation indicates that an event is a direct result of occurrence of another event. A change in one

between two variables. Two variables are related or associated.	variable causes a change in another.
Two variable occur jointly.	Two variables not only occur jointly.
Correlation does not always imply causation.	Causation always implies a correlation between variables.
Does not prove cause and effect.	Establishes a cause-and-effect relationship.
Can occur because of coincidence or a third variable.	Requires evidence that the relationship is causal
Types of correlation include positive, negative and no correlation.	Types of causation include direct causation, indirect causation, bidirectional causation,
Example: Ice-cream sales increase in summer. As temperature rises, (cause) directly lead to increases in ice cream sales (effect).	Example: Increasing the amount of fertilizer causes plant growth to increase, under controlled conditions. However, plant growth might be influenced by additional factors such as sunlight.

42. How is a confidence interval defined in statistics?

Confidence intervals are a range of values within which we can be confident that the true population parameter lies. This range is estimated based on a sample from the population and a chosen level of confidence. The level of confidence speaks to the likelihood that the genuine populace parameter lies inside the certainty interim.

$$Confidence\ Interval = sample\ mean \pm (critical\ value)(standard\ error)$$

where,

The critical value is based on the level of certainty and the degrees of freedom (df) from a t-distribution or z-distribution.

Degrees of freedom(df) = n-1 for a test of measure n.

Standard Error = σ/ √(n).

43. What does the confidence level represent in confidence interval?

44. What is hypothesis testing in statistics?

Hypothesis testing is a statistical method that is used to determine whether there is enough evidence to support or reject a claim about a population based on sample data. Hypothesis testing involves,

- **Defining Hypotheses:**

  Null hypothesis (H_0): Assumes no relationship between variables.

  Alternative hypothesis (H_a): Assumes relationship between variables.

- **Choose significance level (α):** Significance level (usually 0.05) is the maximum chance to accept or reject the null hypothesis.
- **Calculate test statistic:** The test statistic measures how much the sample data deviates from what is expected if the null hypothesis were true. It is a number that helps us decide if the data supports or rejects the claim. Different statistical tests include Z-test, T-test and Chi-square test.
- **Decision:** Test statistic is compared with the critical value or p-value to arrive at a decision.

  Decision based on χ^2_critical value,

  If $\chi^2 > \chi^2$_critical: Reject H_o (null hypothesis)

  If $\chi^2 \leq \chi^2$_critical: Fail to reject H_o (null hypothesis)

  Decision based on p-value for significance level α (typically 0.05),

  If p-value $< \alpha$: Reject H_o (null hypothesis)

  If p-value $\geq \alpha$: Fail to reject H_o (null hypothesis)

45. What is the purpose of a null hypothesis in hypothesis testing?

46. What is the difference between one-tailed and two-tailed test?

One-Tailed Test	**Two-Tailed Test**
One-Tailed test is based on uni-directional hypotheses.	Two-Tailed test is based on nondirectional hypotheses.
It identifies the relation between the variables in one direction (Left or Right direction).	It identifies the relation between the variables in any direction (Left and Right direction).

It determines whether a particular population parameter is larger or smaller than the predefined parameter.	It determines whether a particular population parameter is larger or smaller than the range of values.
It uses one critical value to test the data.	It uses two critical value to test the data.
For one-tailed, hypothesis testing is done by comparing the test static using > or < sign. **Null Hypothesis (H_0):** θ represents a parameter (population mean) and θ_0 is a specific value. **Alternative Hypothesis (H_1):** For a right-tailed test: $H_1: \theta > \theta_0$ For a left-tailed test: $H_1: \theta < \theta_0$	For two-tailed, hypothesis testing is done by comparing the test static using ≠ sign. **Null Hypothesis (H_0):** θ represents a parameter (population mean) and θ_0 is a specific value. **Alternative Hypothesis (H_1):** $\theta \neq \theta_0$.
Rejection region is either from the left side or right side of the sampling distribution.	Rejection region is from both sides i.e. left and right of the sampling distribution.
Critical region lies entirely on either the right side or left side of the sampling distribution.	Critical region is given by the portion of the area lying in both the tails of the probability curve of the test statistic.
Level of significance (α) i.e. 5% has either in the left tail or right tail.	It splits the level of significance (α) into half.

47. What is experiment design and why is it important?

48. What are the key elements to consider when designing an experiment?

49. How can sample size determination affect experiment design?

50. What are some strategies to mitigate potential sources of bias in experiment design?

51. What is the geometric interpretation of the dot product?

52. What is the geometric interpretation of the cross product?

53. How are optimization algorithms with calculus used in training deep learning models?

54. What are observational and experimental data in statistics?

55. How are confidence tests and hypothesis tests similar? How are they different?

56. What is the left-skewed distribution and the right-skewed distribution?

Skewness measures the asymmetry of the distribution of data points around the mean. It is used for identifying any outliers or unusual patterns in the data. Skewness indicates whether the data is concentrated on one side.

Right-skewed distribution (Positive Skewness): Right-skewed distribution is when the mean>median>mode. The outliers are skewed to the right. It has a longer tail on the right side.

Left-skewed distribution (Negative Skewness): Left-skewed distribution is when the mean<median<mode. The outliers are skewed to the left. It has a longer tail on the left side.

57. What is Bessel's correction?

Bessel's correction is the use of n−1 instead of n in the denominator when calculating the sample variance or sample standard deviation. The sample variance is calculated as:

$$s^2 = \frac{\sum_{i=1}^{n}(x - \bar{x})^2}{n - 1}$$

where, s^2 is the sample variance, $\bar{x}$ is the sample mean, n is the sample size.

When we calculate variance from a sample, using n tends to underestimate the true population variance. Using n−1 corrects this bias and gives a better estimate of the population variance. The n−1 is also related to the fact that the sample mean is estimated from the same data, leaving n−1 degrees of freedom.

58. What is kurtosis?

Kurtosis measures the degree of peakedness of a distribution. Kurtosis quantifies the degree of peakedness or flatness in the tails of a probability distribution compared to the normal distribution. A high kurtosis indicates that the distribution has a sharper peak and heavier tails, while a low kurtosis indicates a flatter peak and lighter tails.

A normal distribution has a kurtosis of 3. There are three main types of kurtosis:

Leptokurtic: Distribution has high degree of peakedness. Data is heavily concentrated around the mean, and the tails are relatively thin.

Mesokurtic: Distribution has moderate degree of peakedness. Data is moderately concentrated around the mean, and the tails are neither too thick nor too thin.

Platykurtic: Distribution has low degree of peakedness. Data is widely spread out, and the tails are relatively thick.

59. What is the probability of throwing a two fair dice when the sum is 5 and 8 ?

When two fair dice is thrown, the possible outcomes that the probability of sum is 5 is (1,4), (2,3), (3,2), (4,1).

$$P(sum = 5) = \frac{4}{36}$$

The possible outcomes that the probability of sum is 8 is (2,6), (3,5), (4,4), (5,3), (6,2).

$$P(sum = 8) = \frac{5}{36}$$

Therefore, the probability that sum is either 5 or 8 is given as,

$$P(sum\ is\ 5\ or\ 8) = \frac{4+5}{36} = \frac{9}{36} = \frac{1}{4}$$

60. What is the difference between Descriptive and Inferential Statistics?

61. Imagine that Jeremy took part in an examination. The test has a mean score of 160, and it has a standard deviation of 15. If Jeremy's z-score is 1.20, what would be his score on the test?

Given:

- Mean μ=160
- Standard deviation σ=15
- z-score z=1.20

$$z = \frac{X - \mu}{\sigma}$$

$$X = \mu + (z \times \sigma)$$

$$X = 160 + (1.20 \times 15)$$

$$X = 160 + 18 = 178$$

Jeremy's score on the test is 178.

62. In an observation, there is a high correlation between the time a person sleeps and the amount of productive work he does. What can be inferred from this?

63. What is the meaning of degrees of freedom (DF) in statistics?

Degrees of Freedom (DOF) are the number of independent values in a data set that are free to vary while satisfying a given condition or constraint. It represents the amount of independent information available for statistical analysis. It is calculated using the formula,

$$Degrees\ of\ freedom\ (DF) = n - 1.$$

Degrees of freedom are widely used in statistical tests such as t-tests, chi-square tests, and F-tests.

- For t-tests, small degrees of freedom (df) produce wider distributions with heavier tails. As the degrees of freedom (df) increase, the distribution becomes narrower and closer to the standard normal distribution. For df ≥ 30, the t-distribution is nearly the same as the standard normal distribution.
- For chi-squared test, when df < 3, the distribution has a backward J shape. When df ≥ 3, the distribution becomes hump-shaped and right-skewed. As the degrees of freedom (df) increase, the distribution becomes more similar to a normal distribution.

64. If there is a 30 percent probability that you will see a supercar in any 20-minute time interval, what is the probability that you see at least one supercar in the period of an hour (60 minutes)?

Assuming each 20-minute interval is independent, 60 minutes contains 3 intervals.

- Probability of seeing a supercar in one interval = 0.30
- Probability of not seeing one = 1−0.30 = 0.7

Probability of seeing no supercars in all 3 intervals:

$$0.70^3=0.343$$

Therefore, probability of seeing at least one supercar P(at least one) = 1−P(none) = 1−0.343 = 0.657

65.7% is the probability of seeing at least one supercar in an hour.

65. What is the empirical rule in statistics?

66. What is the relationship between sample size and power in hypothesis testing?

Power measures the ability of a statistical test to detect a true effect when it exists. A high power means the probability of detecting a true effect is high, while a low power means the probability of detecting a true effect is low. The power of a statistical test depends on several factors, including:

1. Sample Size: The bigger the sample size, the higher the power of the test.
2. Effect Size: The bigger the impact estimate, the higher the control of the test.

The calculation of sample size and power depends on the type of statistical test being used. For a t-test, the sample size can be calculated using the following formula:

$$n = ((Z\alpha/2 + Z\beta)^2 * \sigma^2) / d^2$$

where, n = sample size,

Zα/2 = the critical value of the standard normal distribution at the desired level of significance,

Zβ = the critical value of the standard normal distribution at the desired power,
σ = standard deviation of the population,
d = minimum detectable difference.

The power of a test can be calculated using the following formula:

$$Power = 1 - \beta$$

where, β = the probability of making a Type II error

67. Can you perform hypothesis testing with non-parametric methods?

68. What factors affect the width of a confidence interval?

Confidence interval is affected by sample size, level of certainty, and inconstancy of the information.

Sample size: The bigger the test measure, the smaller the confidence interval and smaller the margin of error.

Level of confidence: The higher the level of confidence, the more extensive the confidence interval and bigger the margin of error. The level of certainty suggests a better degree of certainty, which increments the measure of the interim.

Variability of the data: The more variable the information, the more extensive the certainty interim and bigger the margin of error. Higher changeability implies that the information focuses are spread out over a more extensive extend, making it more troublesome to gauge the population parameter.

69. How does increasing the confidence interval affect the width of the confidence interval?

Confidence interval is affected by sample size, level of certainty, and inconstancy of the information.

Level of confidence: The higher the level of confidence, the more extensive the confidence interval and bigger the margin of error. The

level of certainty suggests a better degree of certainty, which increments the measure of the interim.

70. Can a confidence interval be used to make a definitive statement about a specific individual in the population?

71. How does sample size influence the width of a confidence interval?

Confidence interval is affected by sample size, level of certainty, and inconstancy of the information.

Sample size: The bigger the test measure, the smaller the confidence interval and littler the margin of error. A bigger test estimate diminishes the impacts of irregular inspecting blunders and increments the accuracy of the gauges.

72. What is the relationship between the margin of error and confidence interval?

Confidence intervals are a range of values within which we can be confident that the true population parameter lies. This range is estimated based on a sample from the population and a chosen level of confidence. The level of confidence speaks to the likelihood that the genuine populace parameter lies inside the certainty interim.

$$Confidence\ Interval = \ sample\ mean\ \pm \text{Margin of Error}$$

$$\text{Margin of Error } = (critical\ value)(standard\ error)$$

where,

The critical value is based on the level of certainty and the degrees of freedom (df) from a t-distribution or z-distribution.

Degrees of freedom(df) = n-1 for a test of measure n.

Standard Error = σ/ √(n).

73. Can two confidence intervals with different widths have the same confidence level?

74. What is a Sampling Error and how can it be reduced?

Sampling error arises as a sample (a subset) is used to make inferences about a larger population, rather than measuring the entire population. Even with statistically sound sampling methods, the specific sample drawn may not perfectly mirror all characteristics of the entire population. This inherent discrepancy is the sampling error. The size and shape of the sample are used to calculate the sampling error rate, which reflects the accuracy of the selection process. The formula to find the sampling error is given as follows:

$$Sampling\ Error\ (SE)\ =\ (1/\sqrt{N})\ 100$$

Where, N is the Sample Size.

Sampling error can be reduced using,

- **Increase Sample Size** - A larger sample size represents the true population and reduces error. The sample should represent all groups fairly.
- **Stratification** - If the population is similar (homogeneous), sampling is representative. For heterogeneous groups, the population is divided into strata (groups with similar traits) and take samples from each group based on its size.

75. What is a Chi-Square test?

Chi-square test compares observed frequencies (from data) with expected frequencies (from hypothesis). It is used to test how well the data fits an expected distribution. Assumptions of the Chi-squared test includes,

1. Random sample: Data randomly collected
2. Independent observations: Each observation independent
3. Expected frequencies:
    - At least 80% of categories have expected frequency ≥ 5
    - No more than 20% of categories have expected frequency < 5
    - No category has expected frequency < 1

The formula for Chi-Square test statistic χ^2 is given by,

$$\chi^2 = \sum_{i=1}^{n} \frac{(O-E)^2}{E}$$

where, O is the observed value

E is the expected value $E = \frac{n}{k}$ (uniform distribution) and $E = n \times p$ (specific distribution)

n – number of categories

df (degrees of freedom) = n - 1

Hypotheses Testing is performed as follows,

- H_o (null hypothesis): The data follow the hypothesized distribution
- H_1 (alternate hypothesis): The data do not follow the hypothesized distribution

Decision based on χ^2_critical value,

If $\chi^2 > \chi^2$_critical: Reject H_o (null hypothesis)

If $\chi^2 \leq \chi^2$_critical: Fail to reject H_o (null hypothesis)

Decision based on p-value for significance level α (typically 0.05),

If p-value < α : Reject H_o (null hypothesis)

If p-value ≥ α : Fail to reject H_o (null hypothesis)

76. What is a t-test?

77. What is the ANOVA test?

78. How is hypothesis testing utilised in A/B testing for marketing campaigns?

79. What is the difference between one-tailed and two-tailed t-tests?

80. What is an inlier?